

Aim

To study AWS IAM and learn how to manage users, groups, and permissions in AWS.

Theory

Amazon Web Services Identity and Access Management (IAM) is a security service used to control access to AWS resources. IAM helps administrators create users and give them permissions according to their job roles.

In IAM, a **user** represents a person who can log in to AWS. Each user can have a username and password. Users can be organized into **groups** so that permissions can be managed easily. Instead of giving permissions to every user separately, permissions are assigned to groups.

IAM uses **policies** to define permissions. Policies decide what actions a user can perform on AWS services. For example, a policy may allow a user to view Amazon S3 buckets or start and stop Amazon EC2 instances. Policies can be:

- **Managed Policies** – Created by AWS and reusable.
- **Inline Policies** – Custom policies attached to a single user or group.

In this lab:

- The **S3-Support** group had read-only access to Amazon S3.
- The **EC2-Support** group had read-only access to Amazon EC2.
- The **EC2-Admin** group had permission to start and stop EC2 instances.

IAM improves cloud security by providing controlled access to AWS services and resources. It follows the principle of giving only the required permissions to users. This helps protect AWS resources from unauthorized access.

Procedure

1. Opened AWS Management Console.
2. Opened IAM service.
3. Viewed existing users and groups.
4. Checked policies attached to groups.
5. Added:
 - user-1 to S3-Support
 - user-2 to EC2-Support
 - user-3 to EC2-Admin
6. Used IAM sign-in URL to log in as different users.
7. Tested access to Amazon S3 and Amazon EC2 services.

8. Verified permissions for each user.

Result

- Users were successfully added to groups.
 - Permissions worked according to assigned policies.
 - Different users had different access levels in AWS services.
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Conclusion

The lab helped understand AWS IAM, users, groups, and policies. It showed how AWS controls access to resources securely using permissions.

1. **What is AWS IAM?**
AWS IAM (Identity and Access Management) is a service used to securely manage access to AWS resources.
2. **What is the main purpose of IAM?**
The main purpose of IAM is to control who can access AWS services and what actions they can perform.
3. **What is an IAM User?**
An IAM User is an individual account created for a person to access AWS resources.
4. **What is an IAM Group?**
An IAM Group is a collection of users with similar permissions.
5. **What is an IAM Policy?**
An IAM Policy is a document that defines permissions for users, groups, or roles.
6. **What are Managed Policies?**
Managed Policies are pre-created policies provided by AWS.
7. **What are Inline Policies?**
Inline Policies are custom policies attached directly to one user or group.
8. **What is the difference between a User and a Group?**
A user is an individual account, while a group contains multiple users with common permissions.
9. **What is Amazon S3?**
Amazon S3 is a storage service used to store files and data in the cloud.
10. **What is Amazon EC2?**
Amazon EC2 is a service used to create and manage virtual servers in AWS.
11. **Which group had access to Amazon S3 in the experiment?**
The S3-Support group had access to Amazon S3.
12. **Which group had read-only access to EC2?**
The EC2-Support group had read-only access to EC2.
13. **Which user had permission to stop EC2 instances?**
user-3 had permission to stop EC2 instances.
14. **Why could user-2 not stop the EC2 instance?**
Because user-2 had only read-only permissions.
15. **What is meant by Read-Only Access?**
Read-only access allows users to view resources but not modify them.
16. **Why are groups useful in IAM?**
Groups make permission management easier by assigning permissions to multiple users at once.
17. **What is the IAM Sign-In URL?**
It is a special URL used by IAM users to log in to the AWS Management Console.

18. What happens if a user does not have permission for a service?

The user receives an “Unauthorized” error message.

19. What is the principle of least privilege?

It means giving users only the permissions they need to perform their work.

20. What did you learn from this experiment?

I learned how to create and manage users, groups, and permissions using AWS IAM.

Cloud Computing Viva Questions with Answers

1. What is cloud computing?

Cloud computing is the delivery of computing services like storage, servers, and software over the internet.

2. What is distributed computing?

Distributed computing is a system where multiple computers work together to complete a task.

3. What is cluster computing?

Cluster computing is a group of connected computers working together as a single system.

4. What is grid computing?

Grid computing uses computers from different locations to solve large problems together.

5. What is virtualization?

Virtualization is the creation of virtual versions of hardware, servers, or storage devices.

6. What is a hypervisor?

A hypervisor is software that creates and manages virtual machines.

7. Name the types of virtualization.

- Server virtualization
- Storage virtualization
- Network virtualization
- Memory virtualization

8. What is storage virtualization?

It combines multiple storage devices into one virtual storage system.

9. What is network virtualization?

It creates virtual networks independent of physical hardware.

10. What is memory virtualization?

It combines memory from different systems into a single virtual memory pool.

Cloud Computing Basics

11. What are the main characteristics of cloud computing?

- On-demand service
- Broad network access
- Resource pooling
- Rapid elasticity
- Measured service

12. What is an on-premise system?

An on-premise system is where hardware and software are managed locally within an organization.

13. Give one advantage of cloud computing.

It reduces hardware cost.

14. Give one disadvantage of cloud computing.

Internet connection is required.

15. What are the benefits of cloud computing?

- Scalability
- Cost saving
- Easy access
- Backup and recovery

16. What are the risks in cloud computing?

- Data loss
- Security issues
- Privacy problems

Cloud Architecture and Models

17. What is cloud architecture?

Cloud architecture is the structure of cloud components like servers, storage, and networks.

18. What is NIST in cloud computing?

NIST provides standards and guidelines for cloud computing.

19. Name the cloud deployment models.

- Public Cloud
- Private Cloud

- Hybrid Cloud
- Community Cloud

20. What is a public cloud?

A cloud service available to everyone over the internet.

21. What is a private cloud?

A cloud used only by one organization.

22. What is a hybrid cloud?

A combination of public and private cloud.

Cloud Service Models

23. Name the cloud service models.

- IaaS
- PaaS
- SaaS

24. What is IaaS?

Infrastructure as a Service provides virtual hardware resources.

25. What is PaaS?

Platform as a Service provides a platform for application development.

26. What is SaaS?

Software as a Service provides software over the internet.

Storage and Resource Management

27. What is Storage as a Service?

It provides cloud-based storage services to users.

28. Name some cloud storage providers.

- Google Drive
- AWS S3
- Microsoft OneDrive

29. What is resource provisioning?

It is the process of allocating cloud resources to users.

30. What is inter-cloud resource management?

It manages resources between different cloud providers.

IAM and Security

31. What is IAM?

IAM stands for Identity and Access Management used for controlling user access.

32. What is SAML?

SAML is a protocol used for secure user authentication.

33. What is cloud security?

Cloud security protects cloud data, applications, and systems.

34. What are cloud security challenges?

- Data breaches
- Unauthorized access
- Malware attacks

35. What is security governance?

It is the management of security policies and procedures.

36. What are security standards?

Rules and guidelines used to maintain system security.

37. Why is cloud security important?

It protects data and services from cyber attacks.

38. What is authentication?

Authentication verifies the identity of a user.

39. What is authorization?

Authorization decides what resources a user can access.

40. What did you learn from cloud computing?

Cloud computing provides scalable, flexible, and cost-effective computing services over the internet.